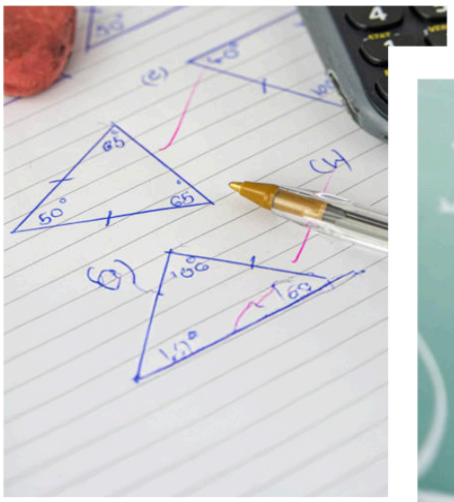
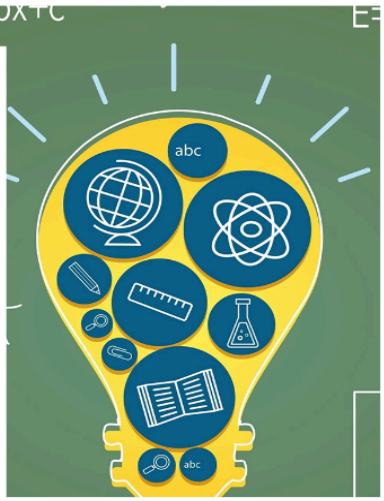
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Co-ordinate Geometry





Coordinate Geometry

Cartesian coordinate system

- The two axes of the coordinate plane are the horizontal x-axis and the vertical y-axis.
- These coordinate axes divide the plane into four quadrants, and the point of intersection of these axes is the origin (0, 0).
- Equation of x-axis : y = 0
- Equation of y-axis : x = 0

Distance between two points

- Distance between a point (x, y) and origin = $\sqrt{x^2 + y^2}$
- Distance between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ = $\sqrt{\left\{\left(x_2 - x_1\right)^2 + \left(y_2 - y_1\right)^2\right\}}$

Midpoint formula: (x, y) = $\left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2}\right)$

Slope formula

- The slope of a line inclined at an angle θ with the positive x-axis is :

  m = Tan θ

- The slope of a line joining the two points (x_1, y_1) and (x_2, y_2) is:

  $m = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)$

- When equation of line is given in the standard form, ax + by + c = 0 is,

  $m = -\frac{coefficient\ of\ y}{coefficient\ of\ x} = -\left(\frac{b}{a}\right)$

Section formula

- **The coordinate of a point which internally divides the line joined by the point (x_1, y_1) and (x_2, y_2) in the ratio m : n is then,**

  $$P(x, y) = \left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$$

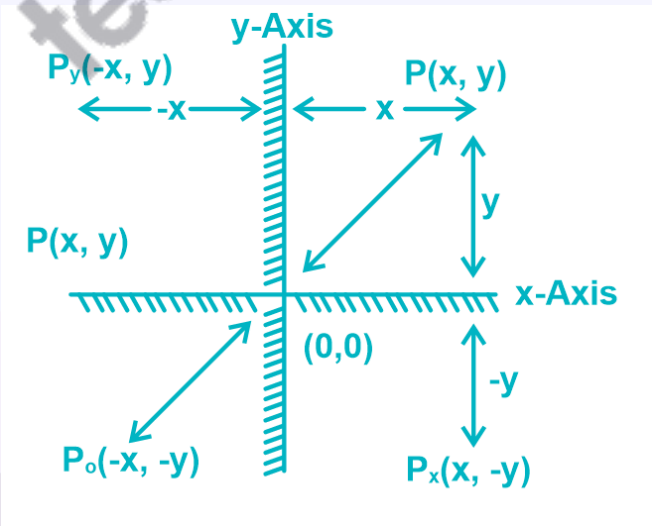
- **If P(x, y) divides the line segment externally in the ratio m:n, then**

  $$P(x, y) = \left(\frac{mx_2-nx_1}{m-n}, \frac{my_2-ny_1}{m-n}\right)$$

Equation of line

- **Slope form :** $y = mx + c$
- **Intercept form :** $\frac{x}{a} + \frac{y}{b} = 1$
- **Point slope form :** $y - y_1 = m(x - x_1)$
- **Two-point form :** $y - y_1 = \left(\frac{y_2-y_1}{x_2-x_1}\right)(x - x_1)$
- **The general form of a line is given by,** $ax + by + c = 0$

Image of a point



- Mirror Image of the point (x, y) about x axis is (x, – y)
- Mirror Image of the point (x, y) about y axis is (- x, y)
- Mirror Image of the point (x, y) about line y = x, is (y, x)
- Mirror Image of the point (x, y) about line y = - x, is (-y, -x)
- Mirror Image of the point (x, y) about origin is (- x, - y)

Angle between 2 lines

- **When slopes of both the lines are given:**

$$\text{Tan } \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

- **When equation of both the lines are given:**

$$\text{Tan } \theta = \left| \frac{b_1 a_2 - a_1 b_2}{a_1 a_2 + b_1 b_2} \right|$$

- **When two lines are parallel to each other:**
  - The angle between them is zero.
  - The slopes of them are equal i.e. $m_1 = m_2$
- **When two lines are perpendicular to each other:**
  - The angle between them is 90 degrees.
  - The product of slopes is always 1 i.e. $m_1 m_2 = -1$
  - When two lines $ax_1+by_1+c = 0$ and $ax_2+by_2+d = 0$ are perpendicular, then : $a_1 a_2 + b_1 b_2 = 0$

Distance between a point and a line

- Perpendicular distance of a line ax + by + c=0 from a point $P(x_1, y_1)$ is

$$D = \frac{|ax_1 + bx_2 + cx_3|}{a^2 + b^2}$$

Shortest distance between two parallel lines

- **If we have the slope-intercept form of the two lines as $y = mx + c_1$ and $y = mx + c_2$, then formula for the distance is:**

$$D = \frac{|C_2 - C_1|}{\sqrt{(1+m^2)}}$$

- **If the equations of the parallel lines are given in the $ax+by+c_1=0$ and $ax+by+c_2=0$, then the formula for the distance is:**

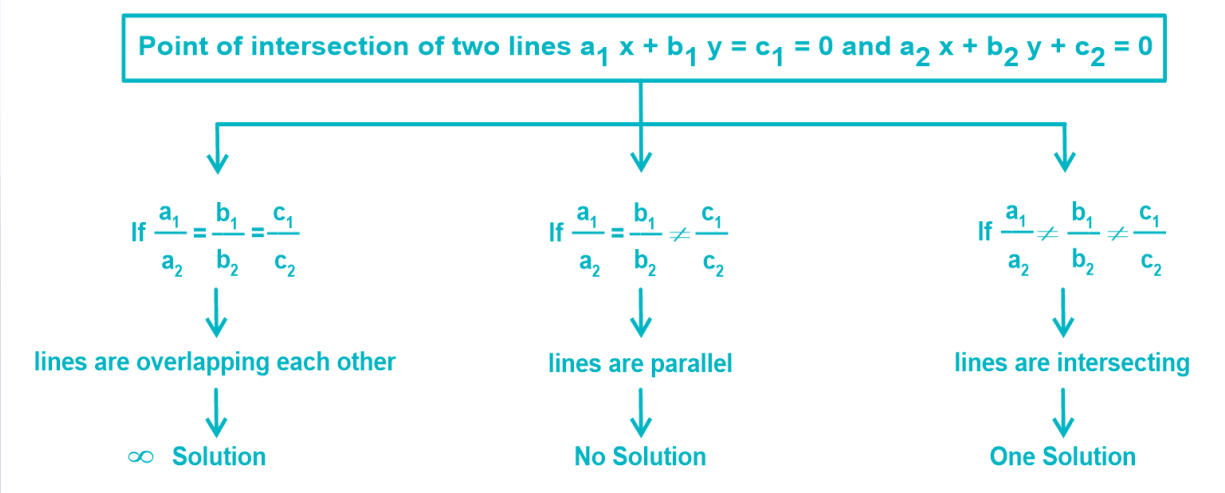
$$D = \frac{|C_2 - C_1|}{\sqrt{(a^2+b^2))}}$$

Coordinate of intersection point of two lines

- **The coordinate of point of intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is:**

$$(x_0, y_0) = \left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right)$$

- **Conditions for point of intersection:**



Area of Triangle

- To find the area of triangle whose vertices are (x_1, y_1) , (x_2, y_2) , (x_3, y_3) :

  Area of triangle = $\frac{1}{2}$ [$x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)$]

- If the area of a triangle is zero, then the three points are collinear.

Centres of Triangle

- **Centroid:**

  Let the coordinates of the vertices of ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3),

  Coordinates of its Centroid = $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

- **Incentre:**

  Let the coordinates of the vertices of ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3),

  Coordinates of its incentre = $\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$

- **Circumcentres:**

  Let the coordinates of the vertices of ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3),

  Coordinates of its circumcentre =

  $$\left(\frac{x_1\sin 2A + x_2\sin 2B + x_3\sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1\sin 2A + y_2\sin 2B + y_3\sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

- **Orthocentre:**

  Let the coordinates of the vertices of ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3),

  Coordinates of its orthocentre =

  $$\left(\frac{x_1\tan 2A + x_2\tan 2B + x_3\tan 2C}{\tan 2A + \tan 2B + \tan 2C}, \frac{y_1\tan 2A + y_2\tan 2B + y_3\tan 2C}{\tan 2A + \tan 2B + \tan 2C}\right)$$